

K33
TUGAS D8 MA1101

No. _____

Date: _____

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Hydarnah

Tutorial 3

25b. $\int \frac{(b+1)\sqrt{b}}{b^2} db$

$$= \int \frac{b\sqrt{b} + \sqrt{b}}{b^2} db$$

$$= \int \frac{1}{\sqrt{b}} + \int \frac{1}{b\sqrt{b}} db$$

$$= \int b^{-\frac{1}{2}} + \int b^{-\frac{3}{2}}$$

$$= (2\sqrt{b} + C_1) + (-2b^{-\frac{1}{2}} + C_2)$$

$$= \boxed{2 \left(\sqrt{b} - \frac{1}{\sqrt{b}} \right) + C}$$

25c. $\int \frac{2}{5} \sec \theta \tan \theta d\theta$ dari $\frac{d}{d\theta} \sec \theta = \tan \theta \sec \theta$

$$= \frac{2}{5} \int \tan \theta \sec \theta d\theta$$

$$= \boxed{\frac{2}{5} \sec \theta + C}$$

25e. $\int x^2 \sqrt{x^3+4} dx$

Misalkan $u = x^3 + 4$

$$\Rightarrow du = 3x^2 dx \Rightarrow dx = \frac{1}{3x^2} du$$

$$= \int x^2 \cdot \sqrt{u} \cdot \frac{1}{3x^2} du$$

$$= \frac{1}{3} \int \sqrt{u} du$$

$$= \frac{2}{9} \sqrt{u^3} + C$$

$$= \boxed{\frac{2}{9} \sqrt{(x^3+4)^3} + C}$$

4

$$25f \int (\tan t + \sec t)^{10} \sec t \, dt$$

$$\text{misalkan } u = \tan t + \sec t$$

$$\Rightarrow du = \sec^2 t + \tan t \sec t \, dt$$

$$\Rightarrow du = \sec t (\tan t + \sec t) \, dt$$

$$\Rightarrow dt = \frac{1}{\sec t \cdot u} du$$

$$\sec t \cdot u$$

$$\int u^{10} \cdot \sec t \cdot \frac{1}{\sec t \cdot u} du$$

$$\sec t \cdot u$$

$$= \int u^9 du$$

$$= \frac{1}{10} u^{10} + C$$

$$= \frac{1}{10} (\tan t + \sec t)^{10} + C$$

5

$$26c \frac{dy}{dx} = \frac{6x - x^3}{2y}$$

$$\Rightarrow 2y \, dy = (6x - x^3) \, dx \quad \text{integralkan}$$

$$\Rightarrow y^2 = 3x^2 - \frac{1}{4}x^4 + C$$

$$(y=3 \text{ dan } x=0)$$

$$\Rightarrow 3^2 = 3(0)^2 - \frac{1}{4}(0)^4 + C \Rightarrow C=9$$

Dengan demikian, persamaan diferensialnya adalah

$$\boxed{y^2 = 3x^2 - \frac{1}{4}x^4 + 9}$$

6

$$27b. \frac{d^2 r}{dt^2} = \frac{2}{t^3}; \quad r'(1) = 1, \quad r(1) = 1$$

Integralkan, di dapat

$$\frac{dr}{dt} = -\frac{1}{t^2} + C \Rightarrow \frac{dr}{dt}(1) = 1 = -\frac{1}{1} + C$$

$$\Rightarrow C=2$$

$$\frac{dr}{dt} = -\frac{1}{t^2} + 2$$

Integralkan lagi, didapat

$$r = \frac{1}{t} + 2t + C \Rightarrow r(1) = 1 = \frac{1}{1} + 2(1) + C$$

$$\Rightarrow C = -2$$

Dengan demikian,

$$\boxed{r = \frac{1}{t} + 2t - 2}$$

31. $h_0 = 10 \text{ m}$, $v_0 = 5 \text{ m/s}$, $g = -10 \text{ m/s}^2$ (arah ke bawah)

$$(a) h = h_0 + v(t) + \frac{1}{2}at$$

$$= 10 + 5t + \frac{1}{2}gt$$

$$\Rightarrow \boxed{h(t) = (10 + 5t - 5t^2) \text{ m}}$$

$$(b) h'(t) = 0$$

$$\Rightarrow 5 - 10t = 0$$

$$\Rightarrow \boxed{t = \frac{1}{2} \text{ sekon}}$$

$$h\left(\frac{1}{2}\right) = 10 + 5\left(\frac{1}{2}\right) - 5\left(\frac{1}{2}\right)^2$$

$$= 10 + \frac{5}{2} - \frac{5}{4}$$

$$= \boxed{11\frac{1}{4} \text{ m}}$$

$$(c) h(t) = 0$$

$$\Rightarrow 10 + 5t - 5t^2 = 0$$

$$\Rightarrow 2 + t - t^2 = 0 \quad [\text{jadi } t^2 - t - 2 = 0]$$

$$\Rightarrow (t-2)(t+1) = 0 \Rightarrow \because t \geq 0, t = 2$$

$$h'(2) = 5 - 10(2) = \boxed{-15 \text{ m/s}}$$

33. Cara Rumus:

$$v_0^2 = v_i^2 - 2as$$

$$\Rightarrow 0 = 20^2 - 2a \cdot 50$$

$$\Rightarrow \boxed{a = 4 \text{ m/s}^2}$$

(atau, agar konsisten,

$$\boxed{a = -4 \text{ m/s}^2})$$

Tutorial 4

$$2c. \int (x^2 + 2x) dx$$

$$\Rightarrow \Delta x = \frac{4-1}{n} = \frac{3}{n}$$

$$x_i = 1 + \Delta x \cdot i$$

$$= 1 + \frac{3i}{n}$$

$$f(x_i) = f\left(1 + \frac{3i}{n}\right)$$

$$= 3 + \frac{12i}{n} + \frac{9i^2}{n^2}$$

$$\Rightarrow \int_1^4 (x^2 + 2x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$$

$$i \Rightarrow n(n+1)$$

$$2$$

$$i^2 \Rightarrow \frac{n(n+1)(2n+1)}{6}$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(3 + \frac{12i}{n} + \frac{9i^2}{n^2}\right) \left(\frac{3}{n}\right)$$

$$= \lim_{n \rightarrow \infty} \frac{3}{n} \sum_{i=1}^n \left(3 + \frac{12i}{n} + \frac{9i^2}{n^2}\right)$$

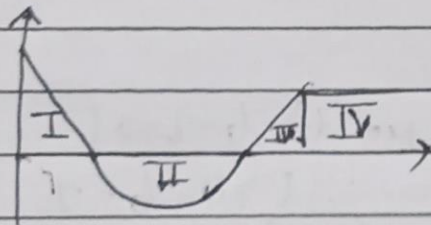
$$= \lim_{n \rightarrow \infty} \frac{3}{n} \left(3n + \frac{12}{n} \left(\frac{n(n+1)}{2}\right) + \frac{9}{n^2} \left(\frac{n(n+1)(2n+1)}{6}\right)\right)$$

$$= 9 + 18 + 9$$

$$= \boxed{36}$$

$$3d. \int_0^6 f(x) \Rightarrow \text{Luas kurva antara } x \in [0, 6]$$

$$\Rightarrow L_1 - L_2 + L_3 + L_4$$



$$L_1 = 1$$

$$L_2 = \frac{1}{2}\pi$$

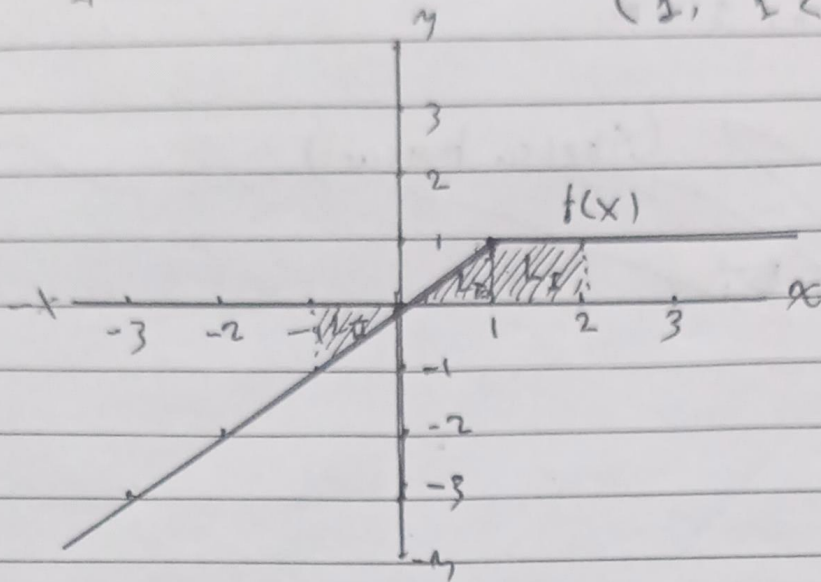
$$L_3 = \frac{1}{2}$$

$$L_4 = 2$$

$$\text{Maka } \int_0^6 f(x) = 1 - \frac{1}{2}\pi + \frac{1}{2} + 2$$

$$= \boxed{\frac{7 - \pi}{2}}$$

4b. $\int_{-1}^2 f(x) dx$ dari $f(x) = \begin{cases} x, & x \leq 1 \\ 1, & 1 < x \end{cases}$



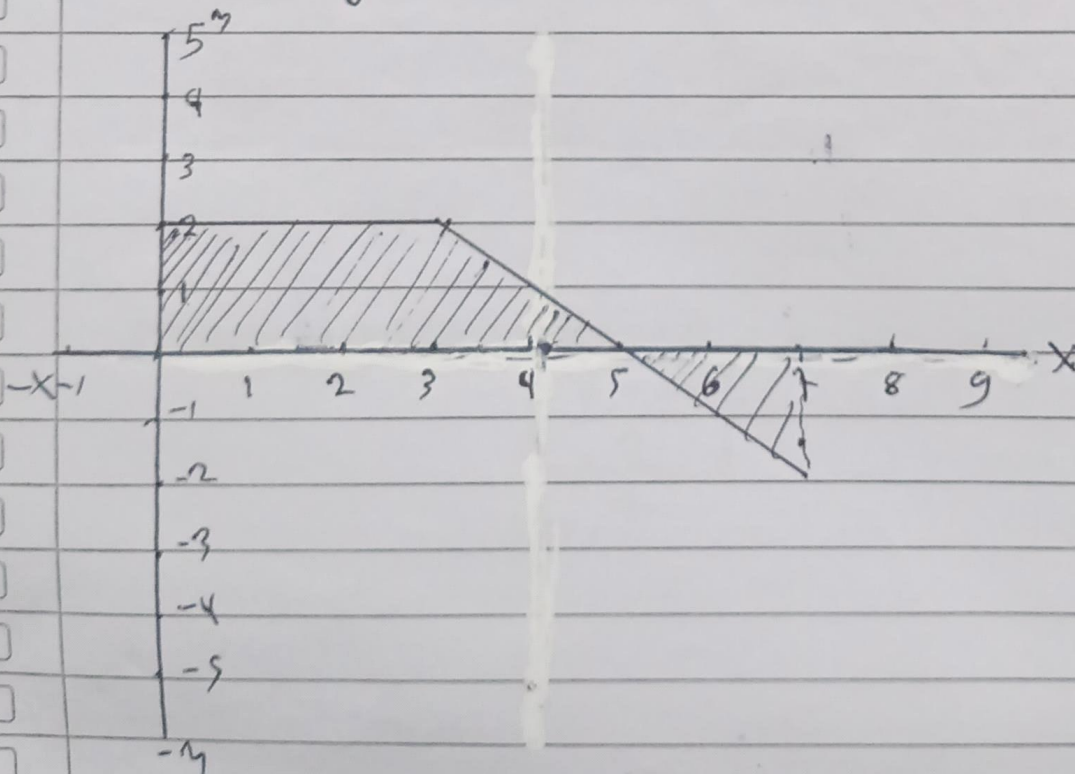
$$L_I + L_{II} - L_{III}$$

$$= \frac{1}{2} + \frac{1}{2} - \frac{1}{2}$$

$$= \frac{1}{2}$$

6. $f(x) = \begin{cases} 2 & 0 \leq x < 3 \\ 5-x & 3 \leq x \leq 7 \end{cases}$

$A(x) = \int_0^x f(t) dt$ dengan $x \in [0, 7]$.



Formula untuk $A(x)$ dapat berupa

$$A(x) = \begin{cases} \int_0^x 2 \, dt & 0 \leq x < 3 \\ A(3) + \int_3^x (5-x) \, dx & 3 \leq x \leq 7 \end{cases} = \begin{cases} 2x - 2(0) \\ 6 + (5x - \frac{1}{2}x^2) - (5(3) - \frac{1}{2}(3)^2) \end{cases}$$

$$= \begin{cases} 2x & 0 \leq x < 3 \\ -\frac{9}{2} + 5x - \frac{1}{2}x^2 & 3 \leq x \leq 7 \end{cases}$$

Dapat dicek bahwa $A'(x) = f(x)$ karena kita hanya menggunakan integral biasa

$$\frac{d}{dx} A(x) = \begin{cases} \frac{d}{dx} 2x & 0 < x < 3 \\ \frac{d}{dx} (-\frac{9}{2} + 5x - \frac{1}{2}x^2) & 3 \leq x \leq 7 \end{cases}$$

$$= f(x) = \begin{cases} 2 & 0 \leq x < 3 \\ 5 - x & 3 \leq x \leq 7 \end{cases}$$

Sesuai dengan Teorema Fundamental Kalkulus

$$\frac{d}{dx} \int_x^{2x+1} \cos(\theta^2) \, d\theta, \text{ misalkan } F(\theta) = \int \cos(\theta^2) \, d\theta$$

$$\Rightarrow \frac{d}{d\theta} F(\theta) = \cos(\theta^2)$$

$$= \frac{d}{dx} F(\theta) \Big|_x^{2x+1}$$

$$= \frac{d}{dx} (F(2x+1) - F(x))$$

$$= 2F'(2x+1) - F'(x)$$

$$= 2 \cos((2x+1)^2) - \cos(x^2)$$

~~Kesimpulan turunan pertama~~

~~$$\frac{d}{dx} F(x) = \frac{d}{dx} \int_0^{x-4} \frac{k}{9+t^2} \, dt$$~~

~~$$= 2t$$~~

~~$$9+t^2$$~~

~~Pembuat nol $2t=0 \Rightarrow t=0$, maka pembagiannya~~

~~$(-\infty, 0)$ dan $(0, \infty)$~~

~~$(-\infty, 0)$ ambil $t=-1 \Rightarrow -\frac{2}{9} < 0$ (turun)~~

~~$(0, \infty)$ ambil $t=1 \Rightarrow \frac{2}{9} > 0$ (naik)~~

$$8. f(x) = \int_0^{2x-4} \frac{1}{4+t^2} dt$$

$$\Rightarrow f'(x) = 2 \left(\frac{(2x-4)}{4+(2x-4)^2} \right)$$

$$= \frac{4x-8}{x^2-4x+5}$$

$$4x^2-16x+20$$

$$= \frac{x-2}{x^2-4x+5}$$

$$x^2-4x+5$$

Kemonotonan ketika pembilang $\rightarrow 0$ sebagai titik absis kritis atau tak terdefinisi

$$\Rightarrow \frac{x-2}{x^2-4x+5} = 0 \Rightarrow x=2$$

$$x^2-4x+5$$

penyebut pasti tak nol karena $D < 0$

$(-\infty, 2)$ ambil $x=0 \Rightarrow -\frac{2}{5} < 0$ (turun)

$(2, \infty)$ ambil $x=3 \Rightarrow \frac{1}{2} > 0$ (naik)

Dengan demikian, selang kemonotonannya adalah

monoton naik = $(2, \infty)$	dengan titik kritis
turun = $(-\infty, 2)$	pada $\boxed{x=2}$

kecekungan menggunakan turunan kedua

$$\Rightarrow f''(x) = \frac{(x^2-4x+5) - (x-2)(2x-4)}{(x^2-4x+5)^2}$$

$$\Rightarrow f''(x) = \frac{-x^2+4x-3}{(x^2-4x+5)^2} \Rightarrow x^2-4x+3=0$$

$$(x^2-4x+5)^2$$

$$(x-1)(x-3)=0$$

maka titik belok absisnya adalah $\boxed{x \in \{1, 3\}}$

Cek kecekungan

$(-\infty, 1)$, ambil $x=0 \Rightarrow \frac{-3}{(0-1)^2} < 0$ (bawah)

$(1, 3)$, ambil $x=2 \Rightarrow \frac{3}{(2-1)^2} > 0$ (atas)

$(3, \infty)$, ambil $x=4 \Rightarrow \frac{-3}{(4-1)^2} < 0$ (bawah)

Dengan demikian, selang kecekungannya adalah

Cekung ke bawah = $(-\infty, 1) \cup (3, \infty)$

Cekung ke atas = $(1, 3)$

9b. $\int_0^{2x} f(t) dt = 8x^2 + \sin(6x)$

$F(2x) - F(0) = 8x^2 + \sin(6x)$ turunkan

$2f(2x) = 16x + 6\cos 6x$

$\Rightarrow f(2x) = 8x + 3\cos 6x$

$\Rightarrow f(x) = 4x + 3\cos 3x$

Dengan demikian,

$f(t) = 4t + 3\cos 3t$

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